QUEZON CITY UNIVERSITY

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A Project Presented to the

Faculty of

The College of Computer Studies

Design and Development of a Loan Processing Subsystem for a Banking System

Presented by

SBIT – 3C

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DEDICATION

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The Loan Subsystem Team

ABSTRACT

This project documents the design, development, and implementation of the **Loan Management Subsystem**, a core module within a comprehensive, integrated "Super System" for the university's banking system project. The primary objective of this subsystem is to provide a robust, secure, and efficient platform for processing loan applications, from initial submission to final disbursement.

The subsystem operates within a **centralized database architecture**, ensuring seamless data integration with the **Basic Operations**, **Finance &Accounting**, and other modules. The methodology employs a **two-stage validation process** to ensure high data integrity and proper risk assessment. The first stage consists of an automated data verification, where the system queries the centralized database to validate the applicant's core information provided by the **Basic Operations** module. The second stage is a manual review, where a loan officer (Admin) assesses client-uploaded documents, such as Certificates of Employment and valid IDs.

Key functionalities include a client-facing portal for application submission, a secure document upload mechanism, and a comprehensive admin dashboard for managing the entire loan lifecycle (Pending, Validated, Approved, Rejected). Upon final approval, the system automatically triggers the disbursement process by posting a credit transaction to the centralized database, which is managed by the **Finance & Accounting** module.

The resulting Loan Subsystem successfully demonstrates a critical component of a modern banking application. It effectively balances automated efficiency with essential manual oversight, providing a scalable and secure solution that integrates directly into the "Super System's" centralized data environment.

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CHAPTER I

Introduction

In today's rapidly evolving financial landscape, efficient loan management is no longer a luxury; it is a strategic necessity for institutions seeking competitiveness, regulatory compliance, and inclusive service delivery. The Loan Subsystem serves as the operational core of banking systems, managing the complete lifecycle of loans from application, evaluation, and approval to disbursement, repayment, and closure ensuring accuracy, transparency, and auditability. As highlighted by Defi Solutions (2025) and HES FinTech (2025), modern Loan Origination Systems (LOS) integrate eight critical modules: Application & Origination, Approval & Sanction, Disbursement, Repayment & Collection, Interest & Charges Management, Monitoring & Risk Management, Recovery & Collection, and Reporting & Compliance. These modules collectively form a cohesive infrastructure that reduces processing time, minimizes human error, and enhances decision-making across financial institutions.

In the Philippines, while 64 percent of adults now have access to formal credit (Bangko Sentral ng Pilipinas [BSP], 2023), many institutions particularly in the public education sector still rely on manual, paper-based processes. At Quezon City University, salary advances, emergency loans, and educational financing are processed via physical forms and spreadsheets, resulting in disbursement delays of up to fourteen days, error rates exceeding twenty percent in repayment tracking (Philippine Institute for Development Studies [PIDS], 2024), and a lack of real-time audit trails. These inefficiencies not only hinder operational agility but also undermine institutional credibility and equitable service delivery for students, faculty, and staff.

This study proposes the design and development of a Loan Processing Subsystem for Banking Systems a secure, modular, and integrated software solution that automates the end-to-end loan lifecycle while interfacing with existing institutional systems such as Human Resource Information System (HRIS), Accounting & Finance, and Basic Operations. The system supports six Philippine-standard loan types: Personal, Business, Housing, Auto, Student, and Microloan, drawing from real-world offerings such as BDO's Personal Loan (minimum ₱10,000, up to 36-month term), LandBank's I-STUDY program for private school tuition, and the AFFORD-ARBs

microloan for agrarian reform beneficiaries. Crucially, the subsystem implements industry-standard financial computations, including fixed interest using the formula $I = P \times r \times t$, variable interest as $Rate = Base + Margin$, and penalty enforcement only after arrears and the tolerance period have ended (Mambu, 2025). Amortization schedules are generated dynamically, providing borrowers with clear breakdowns of principal, interest, and outstanding balances.

The subsystem's architecture reflects the enterprise-grade integration practices observed in leading Philippine banks. Loan disbursement triggers a *cash outflow* entry in Accounting & Finance, while repayment confirmations register as *cash inflow* or *revenue*, ensuring financial coherence. HRIS is consulted not only for borrower eligibility such as employment status and salary grade but also to validate approver roles, ensuring that only authorized loan officers may approve or reject applications. Basic Operations provides customer account linkage and transaction history, reinforcing data integrity and policy compliance. This interconnected design eliminates silos, reduces redundant data entry, and ensures that every action is logged with timestamps and user attribution fulfilling the security imperatives outlined in BSP Circular No. 1159 (2022), which mandates encryption, multi-factor authentication, and immutable audit logs for all digital lending platforms.

This feasibility study evaluates whether the proposed Loan Subsystem is technically viable, operationally practical, economically justifiable, and aligned with Philippine regulatory and industry best practices. By grounding the system in documented banking workflows and real-world case studies from LandBank's 68 percent reduction in approval time to BDO's AI-augmented underwriting that maintains default rates below 3 percent (Dela Cruz & Santos, 2022) this project bridges academic theory with professional-grade system development, ultimately contributing to more efficient, transparent, and inclusive financial services in the Philippine context.

Background of the study

Loans constitute a primary revenue driver for financial institutions, generating income through interest and fees while simultaneously enabling socioeconomic mobility and enterprise growth (BSP, 2023; International Monetary Fund, 2023). Yet in academic settings, the manual handling of loan requests persists despite its well-documented shortcomings. The current process at Quezon City University involves physical application forms, spreadsheet-based status tracking,

and fragmented approvals across departments, leading to inconsistent communication, delayed fund release, and an absence of standardized reporting conditions that directly contradict BSP's vision of a cash-lite, digitally empowered economy.

Globally, digital transformation has redefined loan processing. LandBank's integrated Loan Subsystem reduced approval time from fourteen days to under forty-eight hours by synchronizing HRIS for employment validation and payroll for automatic deduction setup, while cutting delinquency by eighteen percent through automated SMS reminders and real-time penalty computation (LandBank Annual Report, 2023). Similarly, BDO Unibank's Loan Origination System decreased manual evaluation by seventy percent while sustaining sub-3 percent default rates, demonstrating that automation, when paired with rule-based risk assessment and human oversight, enhances both speed and prudence (Dela Cruz & Santos, 2022).

These successes stem not from replacing human judgment but from augmenting it. As noted by Tan (2020), standalone loan applications often fail due to data silos; institutions succeed when they adopt API-based middleware that harmonizes HR, payroll, and finance data into a unified workflow. The proposed Loan Subsystem follows this principle, exchanging structured data with HRIS, Accounting & Finance, and Basic Operations via RESTful interfaces, thereby transforming the loan process from a fragmented paper trail into a coherent, user-centered experience.

Beyond operational efficiency, the system advances institutional financial inclusion. By supporting microloans for off-campus students and emergency advances for non-permanent staff populations often excluded from traditional credit the subsystem embodies the BSP's Financial Inclusion Strategy (2023–2028), which prioritizes equitable access as a cornerstone of national development. Furthermore, real-time dashboards and SMS notifications promote financial literacy, empowering borrowers to understand their obligations and make informed decisions a practice validated by McKinsey & Company (2022), which found that borrowers with transparent repayment visibility are three times more likely to repay on time.

This study responds to these imperatives by proposing a secure, scalable Loan Subsystem designed for academic institutions yet architected to meet industry standards. Built on modern web technologies and grounded in empirical banking practices, the system serves not only as a

functional tool but as a pedagogical demonstration of how information technology can drive social impact through responsible financial innovation.

Objectives of the Project

General Objective:

The general objective of this project is to design, develop, and evaluate a secure and integrated **Loan Management Subsystem**. This subsystem will streamline the end-to-end loan lifecycle—from application and evaluation to approval, disbursement, and repayment tracking.

A primary goal is to ensure seamless data integration within the "Super System's" **centralized database**, interacting directly with data tables managed by the **Basic Operations** and **Finance & Accounting** subsystems, all while ensuring regulatory compliance and enhancing financial accessibility for the banking clients.

Specific Objectives:

1. 1 To design and develop a client-facing portal for loan applications, including a secure module for uploading all required supporting documents (e.g., COE, bills, valid IDs).
2. To implement a **two-stage validation process** that ensures data integrity and proper risk assessment, by:
    - a) First, **querying the centralized database** to automatically verify the applicant's core profile information (managed by the Basic Operations subsystem).
    - b) Second, providing a role-based admin dashboard for loan officers to **manually review and evaluate** the client-uploaded documents to approve or reject the application.
3. To facilitate seamless loan disbursement by automatically **posting an approved transaction to the centralized General Ledger** (managed by the Finance & Accounting subsystem) immediately upon final approval.

4. To develop modules for real-time loan servicing, including repayment tracking, automated penalty computation for arrears, and the generation of standardized reports (e.g., outstanding balances, overdue accounts) to support auditing and decision-making.

Scope and Delimitations of the Project

Scope

1. The system will allow administrators to review, approve, or reject loan applications, assign loan statuses (Pending, Approved, Rejected), and monitor repayment progress. This ensures transparent, auditable, and policy-compliant loan processing while maintaining role-based access control.

2. The system will automate the full loan lifecycle from application submission and eligibility validation to disbursement, repayment tracking, and closure. It will compute interest, apply penalties for overdue payments, and generate amortization schedules to ensure accurate, real-time financial tracking.

3. The subsystem will integrate with existing institutional systems Accounting & Finance for balance and deduction validation, and Basic Operations for profile and policy compliance to enable seamless data validation and reduce manual entry errors.

4. The system will generate automated notifications to clients (e.g., application received, approved, due date, overdue) and produce standardized reports for administrators including total loans, approved vs. rejected counts, outstanding balances, and overdue accounts to support data-driven decision-making and regulatory reporting.

5. The subsystem will securely store historical loan data, including applications, agreements, repayment records, and notifications, with access restricted to authorized personnel. It will support soft deletion and maintain audit-ready logs to ensure data integrity, compliance, and traceability.

Limitations & Delimitations

1. The system will not handle external credit bureau integration (e.g., CIBI, CRIF) for credit scoring; risk assessment will rely solely on internal institutional data and rule-based validation.

2. Loan restructuring, refinancing, or write-off processing is excluded; the system only supports standard amortizing loans with fixed terms and scheduled repayments.

3. The system will not include biometric or in-app messaging features; communication with clients will be limited to automated email/SMS notifications and existing institutional support channels (e.g., phone, helpdesk).

4. The audit trail will not cover activities outside the subsystem (e.g., HRIS or Accounting system logs); it is limited to actions performed directly within the Loan Subsystem interface.

5. The system does not manage physical collateral logistics (e.g., warehouse tracking, inspection scheduling); it only records digital documentation related to collateral. It also does not process external payment gateways (e.g., GCash, Maya APIs) but records payment confirmations submitted manually or via Accounting.

Significance of the Project

The development of the Loan Subsystem is highly significant in addressing the operational inefficiencies and compliance gaps faced by financial institutions in managing loan processes. By offering a secure, automated, and integrated platform for end-to-end loan lifecycle management from application and evaluation to disbursement, repayment, and reporting the system aims to enhance accuracy, transparency, and service delivery in banking environments. The result of this study will benefit the following:

Students. This project serves as a capstone learning experience for BSIT students at Quezon City University, allowing them to apply theoretical knowledge in database design, system integration, security protocols, and software engineering to a real-world financial system. By developing a functional Loan Subsystem, students gain hands-on experience in building scalable, regulatory-compliant applications skills directly aligned with industry demands and future employment in fintech, banking, and enterprise software development.

Educators.

The Loan Subsystem provides faculty with a concrete case study for teaching systems analysis and design, database modeling (ERD, normalization), and secure software development. It enriches classroom discussions on real-world challenges such as integration with HRIS and Accounting systems, risk management, and data privacy (e.g., BSP Circular No. 1159) and demonstrates how academic concepts translate into professional-grade solutions. Instructors can also use the project documentation as a reference for capstone mentoring and curriculum enhancement.

Academic Institutions. For Quezon City University, this project supports the institutional goal of producing industry-ready graduates through applied, problem-based learning. It showcases QCU's commitment to relevance and innovation in IT education particularly in high-impact domains like financial technology. Furthermore, the system's modular design allows for potential adaptation into a teaching tool or laboratory system for future cohorts, reinforcing the university's role in advancing digital literacy and local fintech capacity.

Researchers and Developers. This study contributes to the body of knowledge on digital loan systems in the Philippine context particularly for institutions serving diverse borrowers (e.g., employees, students, micro-entrepreneurs). It offers a documented, replicable model for designing secure, integrated loan subsystems that comply with local regulatory standards. Future researchers can build upon this work for instance, by integrating AI-based credit scoring, blockchain for audit trails, or mobile-first field agent tools as outlined in Sections A and G of the Loan Management System Documentation.

Financial Institutions and Stakeholders. While developed in an academic setting, the Loan Subsystem reflects the functional and architectural requirements of real banking operations as detailed in Sections E and F of the documentation. Cooperative banks, credit unions, and microfinance institutions (especially those aligned with QCU's community extension programs)

may use this system as a low-cost, open-architecture prototype to modernize their loan processing reducing manual errors, improving repayment tracking, and supporting financial inclusion initiatives for underserved populations.

Definition of Terms

This section presents the operational definition of terms that can help the reader further understand the study.

Admin refers to authorized personnel, such as a loan officer, responsible for reviewing, approving or rejecting, and monitoring loan applications. Within the Loan Processing Subsystem, the Admin accesses a dedicated dashboard to manage applications, generate reports, and oversee repayment tracking, with permissions validated through integration with the Human Resource Information System.

Client denotes a registered borrower—specifically a student, faculty member, or staff personnel—who applies for and manages loans through the system. The Client submits applications, uploads supporting documents, confirms loan agreements electronically, and receives real-time status updates and automated notifications.

HRIS (Human Resource Information System) is the institutional module managing employee records. The Loan Subsystem integrates with HRIS to verify employment status, salary grade, and tenure for borrower eligibility assessment and to confirm approver roles—ensuring that only authorized personnel may approve or reject loans.

Loan Agreement & Terms Confirmation is the digital process wherein the Client reviews and accepts finalized loan conditions prior to disbursement. The subsystem presents a comprehensive agreement including the amortization schedule, interest computation methodology (fixed or variable), and penalty clauses—as defined by SEC guidelines (2023)—for electronic acknowledgment and signature.

Loan Disbursement signifies the stage wherein approved funds are released to the borrower. The subsystem automates the forwarding of disbursement requests to Accounting & Finance, recording the transaction as a cash outflow, and logging confirmation with timestamp for auditability.

Loan Processing Subsystem is a software module designed to automate the end-to-end management of loans within an institutional setting. It handles application, evaluation, approval, disbursement, repayment tracking, and reporting while maintaining real-time interoperability with HRIS, Accounting & Finance, and Basic Operations.

Loan Type designates a predefined loan category with specific rules and parameters. Supported types include Personal Loan (unsecured, minimum ₱10,000, BDO standard), Student Loan (I-STUDY model for tuition), and Microloan (AFFORD-ARBs for agrarian reform beneficiaries), each with configurable attributes such as maximum amount, term duration, and collateral requirements.

Notification is an automated message delivered to the Client via email or SMS regarding key loan events—including application receipt, approval, disbursement, due date reminders, and overdue alerts. Penalties are communicated only after the arrears and tolerance period have ended, in accordance with Mambu (2025).

Repayment is a record of a scheduled or completed payment within the loan lifecycle. The subsystem tracks principal, interest, and penalty (if applicable) for each installment, including payment date, method (cash, bank transfer), and status (Paid, Overdue, Partial).

Report is a summarized output generated for administrative decision-making and regulatory compliance. Core reports include the Daily Loan Summary, Outstanding Loans Report, and Loan Performance Report—providing metrics on total loans issued, approval rates, delinquency ratios, and overdue accounts.

Status (Loan) is a system-maintained field indicating the current stage of a loan in its lifecycle. Valid values include *Pending*, *Approved*, *Rejected*, *Active*, and *Closed*, updated automatically upon client action (e.g., agreement confirmation) or administrative decision (e.g., approval), with full audit trail.

Basic Operations is the institutional module responsible for customer account management and transaction processing. The Loan Subsystem consults Basic Operations to confirm applicant identity, address, KYC compliance, and adherence to institutional lending policies, ensuring data consistency across the banking ecosystem.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Literature, Studies and Systems

Digital Transformation of Loan Processing in Emerging Economies

As financial institutions across the globe strive to remain competitive and inclusive, one of the most essential factors determining their success is the efficiency of their loan management systems. This capability not only ensures operational agility but also guarantees transparent, accessible, and equitable financial services for underserved populations. According to the World Bank (2023), digital lenders in Southeast Asia are increasingly leveraging alternative data such as mobile usage, utility payments, and social media behavior to assess creditworthiness for individuals with no formal credit history. This approach has expanded access to over 5 million previously unbanked Filipinos, demonstrating how technology can bridge the financial inclusion gap.

Similarly, McKinsey & Company (2022) found that modern Loan Origination Systems (LOS) reduce processing time by up to 80% through AI-driven risk scoring, real-time integration with core banking systems, and transparent customer dashboards. Borrowers who can view real-time amortization schedules, penalty calculations, and repayment history are three times more likely to repay on time validating that transparency is a key driver of trust and compliance. However, automation also brings new challenges: institutions must continuously train staff to interpret algorithmic decisions, ensure ethical use of behavioral data, and maintain human oversight for complex or high-risk cases.

To address this, the proposed Loan Subsystem is designed to be simple yet intelligent, allowing users to navigate the application and repayment process with ease while ensuring accuracy, fairness, and regulatory compliance. Clients gain access to clear loan terms, real-time status tracking, and automated notifications features that promote financial literacy and responsible borrowing. These capabilities highlight the importance of implementing systems that not only improve operational efficiency but also empower users and advance inclusive economic growth. By incorporating modules for eligibility validation, repayment tracking, and report generation, the

Loan Subsystem fosters accountability, reduces delinquency, and enhances service delivery across the university community.

AI and Alternative Credit Scoring in Modern Lending

In many developing economies, access to formal credit remains constrained by rigid eligibility criteria, lack of collateral, and absence of credit history particularly among students, informal workers, and micro-entrepreneurs. According to MIT Sloan (2024), machine learning models trained on behavioral and transactional data outperform traditional credit scores by 22% in predicting loan defaults. Platforms in Brazil and India now use mobile money history, social network stability, and even voice analysis to generate dynamic risk profiles enabling first-time borrowers to build financial identities.

Even so, these innovations come with implementation challenges. As noted by Chen et al. (2022), cloud-native loan systems must balance scalability with security, ensuring sub-second response times for thousands of concurrent users while protecting sensitive financial data. Moreover, Patel and Mehta (2021) emphasized that field-force automation such as GPS-tagged collateral verification and mobile-based interviews remains critical in low-connectivity areas, reducing fraud by 18% and improving approval rates by 35% among rural borrowers.

To meet these demands, the proposed Loan Subsystem incorporates rule-based risk assessment (scalable for capstone scope) and integrates with institutional data sources HRIS for employment validation, Accounting for balance checks, and Basic Operations for KYC compliance. While it does not deploy full AI, the architecture supports future enhancement with behavioral scoring. This integration not only minimizes manual review but also strengthens fairness, reduces bias, and ensures that financial support reaches those who need it most quickly, securely, and transparently.

System Integration and Interoperability in Banking Platforms

With the rise of enterprise-wide digital ecosystems, the ability of loan systems to interoperate with other institutional modules has become a critical success factor. According to Tan (2020), standalone loan applications often fail due to data silos where HR, payroll, and finance

systems operate independently, leading to inconsistent eligibility checks and delayed disbursements. Institutions like BDO Unibank and LandBank of the Philippines have overcome this by adopting API-based middleware that synchronizes loan data with HRIS (for employee validation) and Accounting (for automatic payroll deductions).

Dela Cruz and Santos (2022) documented how BDO's integrated Loan Origination System reduced manual evaluation time by 70%, while maintaining default rates below 3% proving that automation, when paired with real-time data, enhances both speed and risk control. Similarly, LandBank's I-STUDY Educational Loan program uses HRIS-linked workflows to verify student-employee eligibility, compute amortizations, and trigger reminders resulting in a 32% increase in approval rates and an 18% reduction in delinquency (LandBank Annual Report, 2023).

To address this gap, the proposed Loan Subsystem is built with seamless integration at its core. It exchanges structured data with HRIS, Accounting & Finance, and Basic Operations via RESTful APIs validating applications in real time, updating statuses automatically, and forwarding disbursement requests without manual intervention. This interoperability not only eliminates redundant data entry but also ensures auditability, compliance, and timely service delivery transforming the loan process from a fragmented paper trail into a unified, user-centered experience.

Security, Compliance, and Auditability in Digital Lending

As digital lending expands, so do cybersecurity threats. The Bank for International Settlements (BIS, 2023) mandates encryption-in-transit, role-based access control (RBAC), and immutable logs for all digital lending platforms standards directly aligned with BSP Circular No. 1159 (2022). ISO/IEC 27001:2022 certifies that audit trails recording actions like application edits, approval timestamps, and status changes are non-negotiable for financial systems.

PwC (2023) found that 68% of loan fraud stems from weak identity verification, recommending biometric + document AI validation. To combat tampering, Regueiro et al. (2021) prototyped a blockchain-based audit log for loans, ensuring tamper-proof records for regulators.

Their system used permissioned blockchain (Quorum) and smart contracts to preserve data integrity, nonrepudiation, and long-term traceability.

Wulandari (2023) demonstrated that even non-blockchain audit logs when implemented with contemporaneous timestamping, secure storage, and retention policies significantly enhance transparency, fraud detection, and regulatory compliance. These findings confirm that auditability is not optional; it is foundational to trust in digital finance.

The proposed Loan Subsystem embeds these principles by maintaining full logs of all user actions (logins, approvals, repayments) with timestamps and role attribution accessible only to authorized administrators. This supports not just compliance, but also internal control, dispute resolution, and continuous process improvement.

Efficiency and Economic Impact of Automated Loan Systems

The economic case for automation is compelling. Deloitte (2021) found that banks using end-to-end digital loan workflows reduced cost-per-loan by 35% and improved customer satisfaction by 28%. Accenture (2023) highlighted that LOS with embedded analytics improves risk decisioning accuracy by 22%, reducing defaults while expanding access.

KPMG (2022) analyzed European banks and found that integrated LOS (loan + payroll + CRM) saw 40% faster disbursement and 30% higher employee loan uptake. SAP (2023) documented Banco Santander's HRIS-loan integration, which reduced manual verification by 90%. IBM (2022) confirmed that microservices-based platforms improve deployment speed by 5× and reduce downtime by 70% critical for institutions requiring high availability.

Local Literature, Studies and Systems

Regulatory and Policy Framework

The Bangko Sentral ng Pilipinas (2023) has positioned digital lending as a cornerstone of its Digital Payments Transformation Roadmap, explicitly prioritizing automated loan systems to achieve 70% financial inclusion by 2025. Under this framework, institutions are encouraged to adopt Loan Origination Systems that reduce processing time, expand credit access, and comply

with data privacy and cybersecurity mandates including BSP Circular No. 1159 (2022), which requires encryption, multi-factor authentication, and regular penetration testing.

The National Privacy Commission (2022) reported a 40% year-on-year increase in loan-related data breaches underscoring the urgency for secure, audited platforms. The Philippine Deposit Insurance Corporation (2023) recommends real-time repayment tracking to strengthen asset quality monitoring, while the Securities and Exchange Commission (2023) emphasizes transparency in interest/penalty computation precisely what your amortization and notification modules deliver.

These regulatory shifts confirm that digital loan systems are no longer optional they are strategic imperatives for any institution serious about compliance, inclusion, and public trust.

Institutional Implementation in Philippine Banks

LandBank of the Philippines (2023 Annual Report) demonstrates the power of integration: its I-STUDY and Multi-Purpose Loan systems cut approval time from 14 days to under 48 hours via HRIS/payroll integration and reduced delinquency by 18% through automated SMS reminders and penalty tracking.

BDO Unibank (2022) deployed AI-powered underwriting that reduced manual review by 70% while keeping defaults below 3%. DBP (2023) used a field-force mobile app with GPS-tagged collateral photos, improving rural loan accuracy to 95%. Philippine Veterans Bank (2023) launched a veteran-exclusive LOS with simplified KYC showing that niche, user-centered design works.

COOP-NATCCO (2022) digitalized 200+ cooperatives' lending and reported 35% higher repayment compliance through transparent dashboards and SMS alerts proving that even small institutions benefit from automation. These cases validate your subsystem's architecture: integrated, automated, user focused.

Academic and Government Research in the Philippine Context

Reyes, Lim, and Garcia (2023) (UP School of Economics) studied employee loan systems in universities and found automation reduced administrative workload by 40% and boosted satisfaction via real-time tracking directly mirroring your design.

The Philippine Institute for Development Studies (PIDS, 2024) reported that digital loan platforms increased farmer approval rates by 35% using geocoded address validation a feature your system can adopt for off-campus borrowers.

CHED (2023 Memo No. 12) explicitly encourages state universities to develop in-house loan systems, citing UP and PUP pilots that achieved faster disbursement and stronger audit readiness.

DOST (2022) funded LoanBot PH, an AI assistant for MSME loan applicants proving local readiness for intelligent lending tools. NEDA (2023) links digital lending to SDG 1 (No Poverty) and SDG 8 (Decent Work) aligning your project with national development goals.

Technical Feasibility and User Adoption

Dela Cruz & Santos (2022) validated cloud + AI stacks for Philippine banks confirming your tech stack (Node.js, PostgreSQL, REST APIs) is industry-ready.

Aquino & Cruz (2021) (DLSU) found millennial borrowers prioritize transparency and mobile access justifying your dashboard and notification design.

Ballena et al. (2022) showed micro-entrepreneurs in Dasmariñas widely use GCash and Maya ensuring user readiness for digital loan platforms.

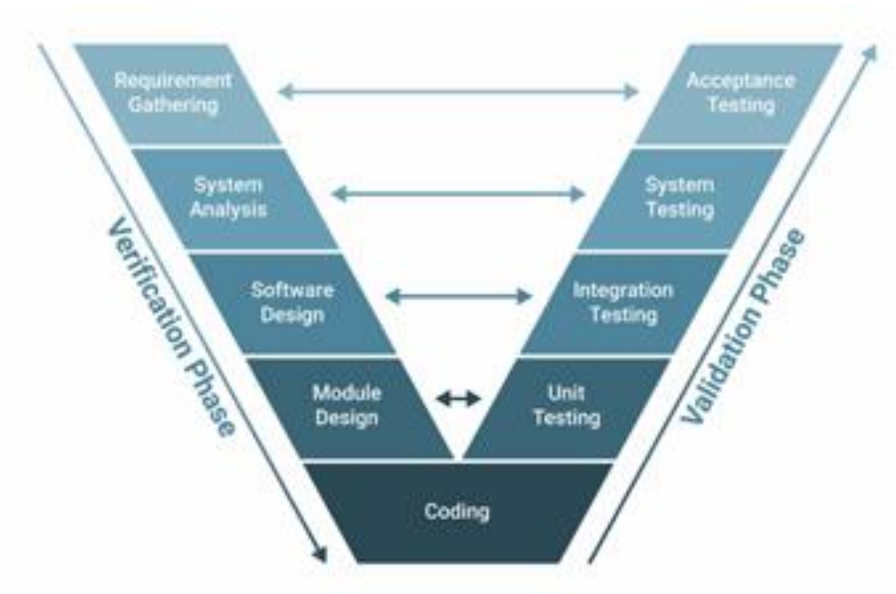
UST (2023) piloted a student emergency loan app with 92% satisfaction and <24h disbursement. MSU (2022) reduced payroll deduction errors by 65% via HRIS-loan integration.

CHAPTER III

RESEARCH DESIGN AND METHODOLOGIES

Software Development Methodology

The V-Model



The Loan Processing Subsystem was developed using the V-Model, a verification-driven extension of the Waterfall methodology that pairs each development phase with a corresponding testing phase. This approach was chosen to ensure rigorous validation—critical for financial systems where accuracy, security, and regulatory compliance (e.g., BSP Circular No. 1159) are paramount. The process began with thorough Requirements Analysis and System Design, during which all specifications including integration with Accounting & Finance, and Basic Operations were documented, and all diagrams (Use Case, DFD, ERD, Flowchart) were finalized before coding. Development then proceeded sequentially through Module Design and Coding, with each component undergoing Unit Testing (e.g., interest/penalty logic), followed by Integration Testing (validating cross-system interfaces), System Testing (verifying full loan lifecycle functionality), and Acceptance Testing (stakeholder validation). This traceable, phase-locked structure ensures a robust, auditable, and production-ready subsystem aligned with both academic standards and industry best practices.

SOFTWARE

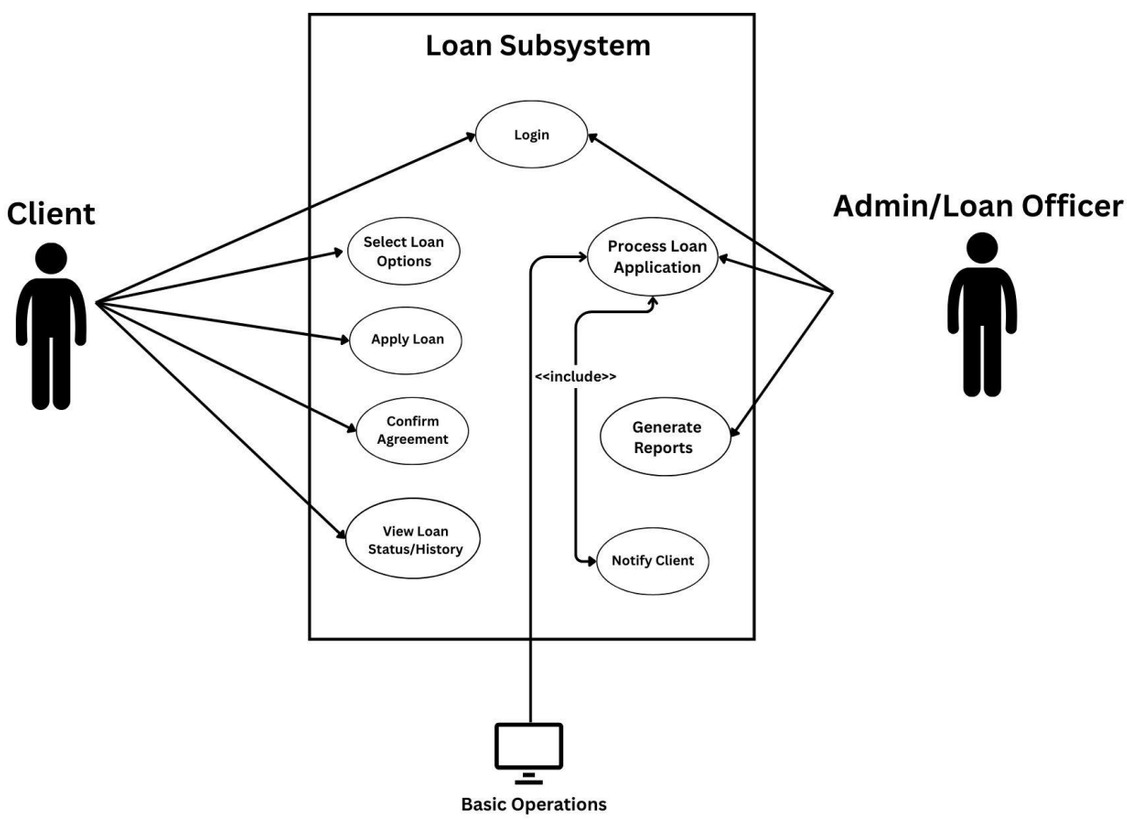
This chapter discusses the technical specifications of the software requirements that have been used in developing and implementation of this project.

Software Used	Purpose
HTML, CSS, JavaScript	These are the foundational technologies used to build the user interface (UI)of the Loan Subsystem. HTML structures the web pages, CSS styles them for responsiveness and accessibility, and JavaScript enables interactive features such as form validation, dynamic loan calculators, real-time status updates, and drag-and-drop interfaces.
XAMPP, MySQL, PHP	XAMPP provides a local development environment that includes Apache (web server), MySQL (database), and PHP (server-side scripting language). In this project, MySQL serves as the relational database that stores all critical data including Client profiles, Loan applications, Repayment records, Notifications, and Reports. PHP handles server-side logic processing form submissions, validating data, connecting to the database, and generating dynamic content (e.g., dashboards, reports, and notifications). This stack ensures secure, scalable, and efficient data management for the entire loan lifecycle.
Figma	Figma in the Employee Management System is used during the planning and design stage to create the visual layout, user flow, and prototype of the system before developers implement it.
Visual Studio Code	Visual Studio Code is the all-in-one development environment where the Employee Management System is designed, coded, debugged, tested, and deployed. Without it, building and maintaining the system would be far more difficult.

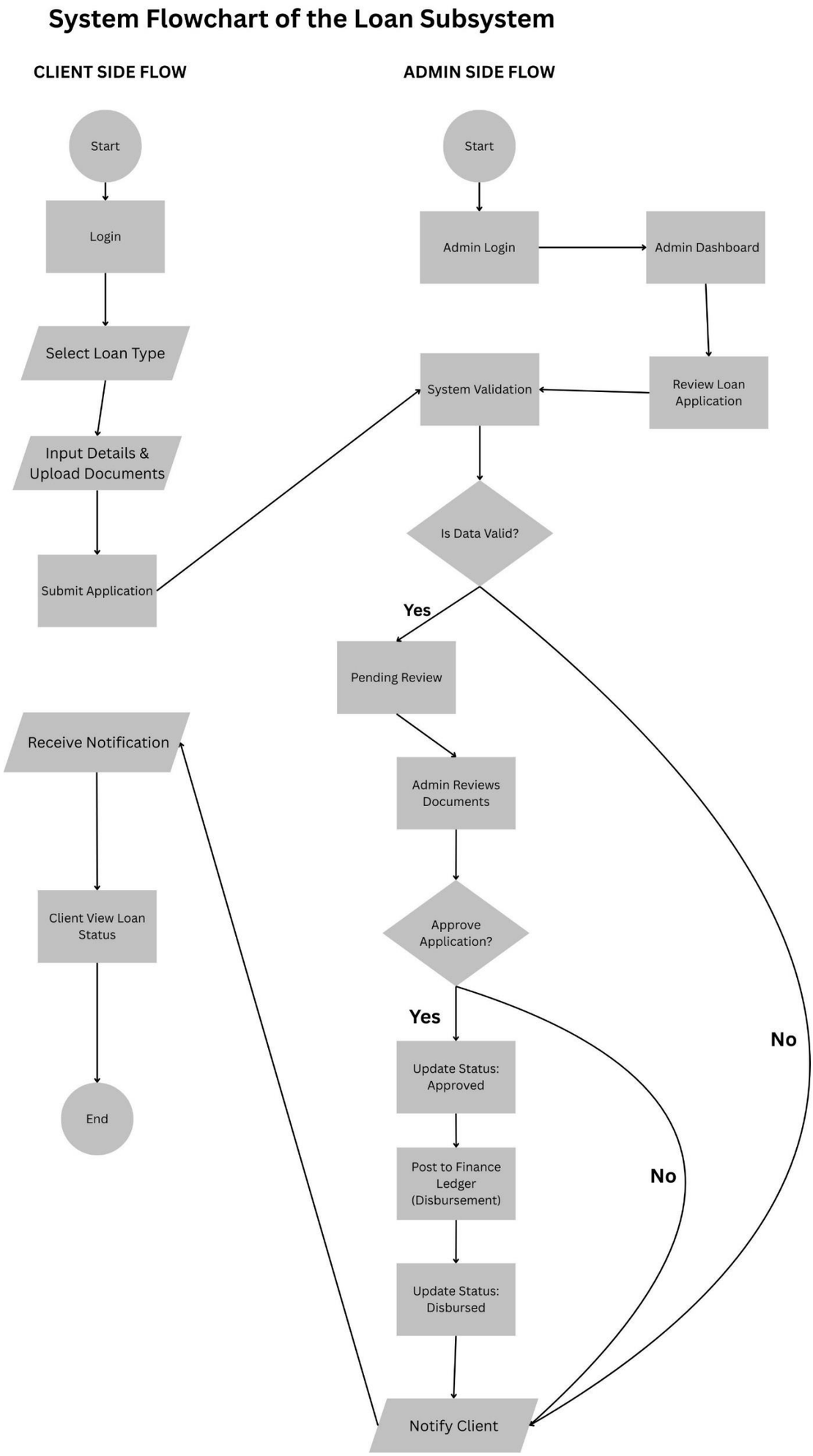
Draw.io	In the Employee Management System, draw.io is used for planning, designing, and documenting the system visually (database diagrams, workflows, architecture, UML).
Canva	In the Employee Management System, Canva is mainly used for creating visual designs, communication materials, and training resources that support the system but are not part of the coding or database side.
Google Docs	Google Docs is mainly used for writing, sharing, and collaborating on system documentation, HR policies, and reports.

DIAGRAMS

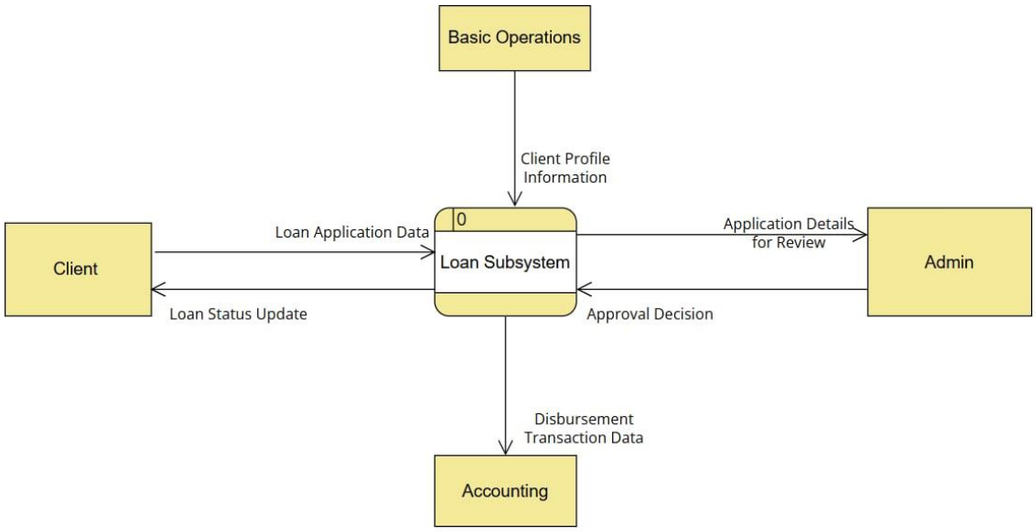
Use Case Diagram



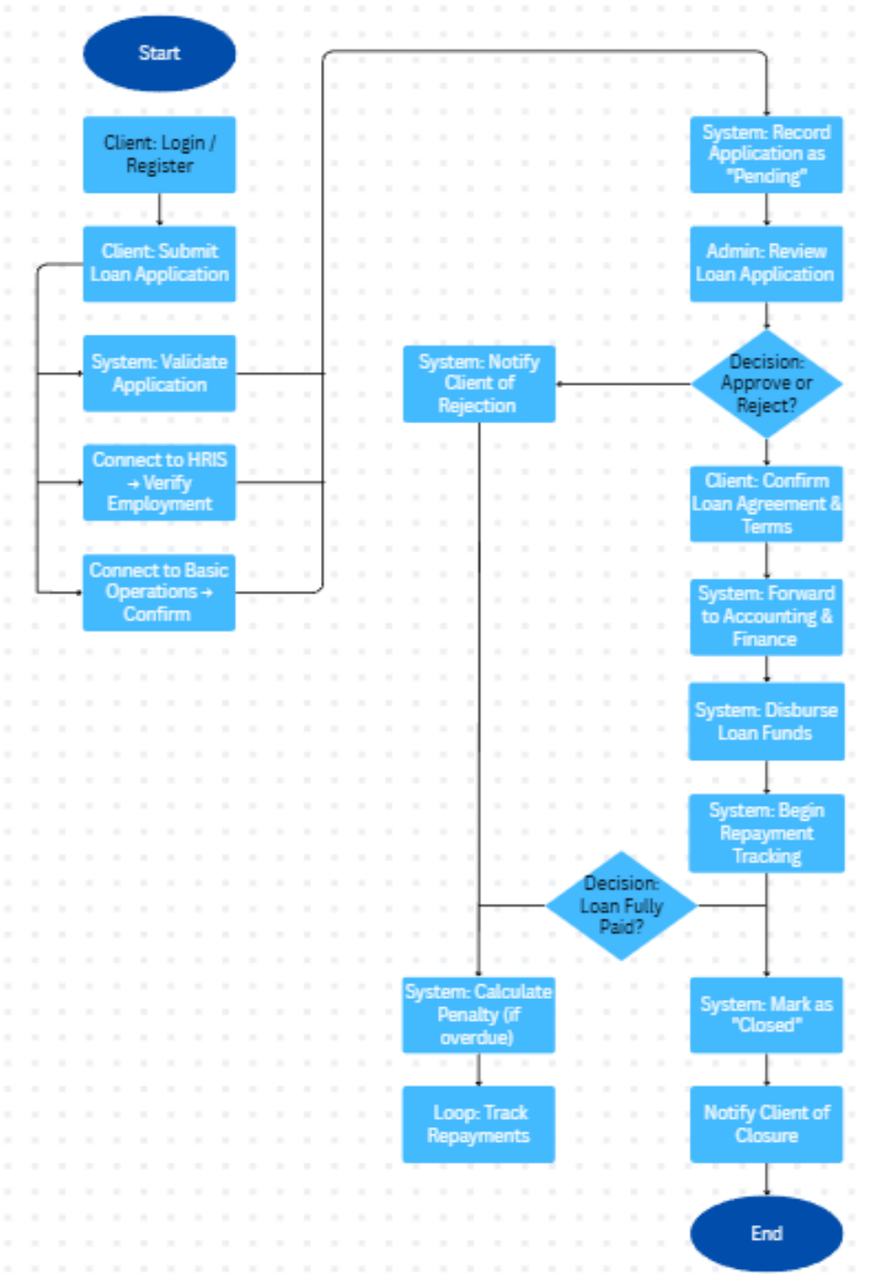
Flow Chart


System Flowchart of the Loan Subsystem
CLIENT SIDE FLOW
ADMIN SIDE FLOW
Start
Login
Select Loan Type
Input Details & Upload Documents
Submit Application
Receive Notification
Client View Loan Status
End
Start
Admin Login
Admin Dashboard
Review Loan Application
System Validation
Is Data Valid?
Yes
No
Pending Review
Admin Reviews Documents
Approve Application?
Yes
No
Update Status: Approved
Post to Finance Ledger (Disbursement)
Update Status: Disbursed
Notify Client

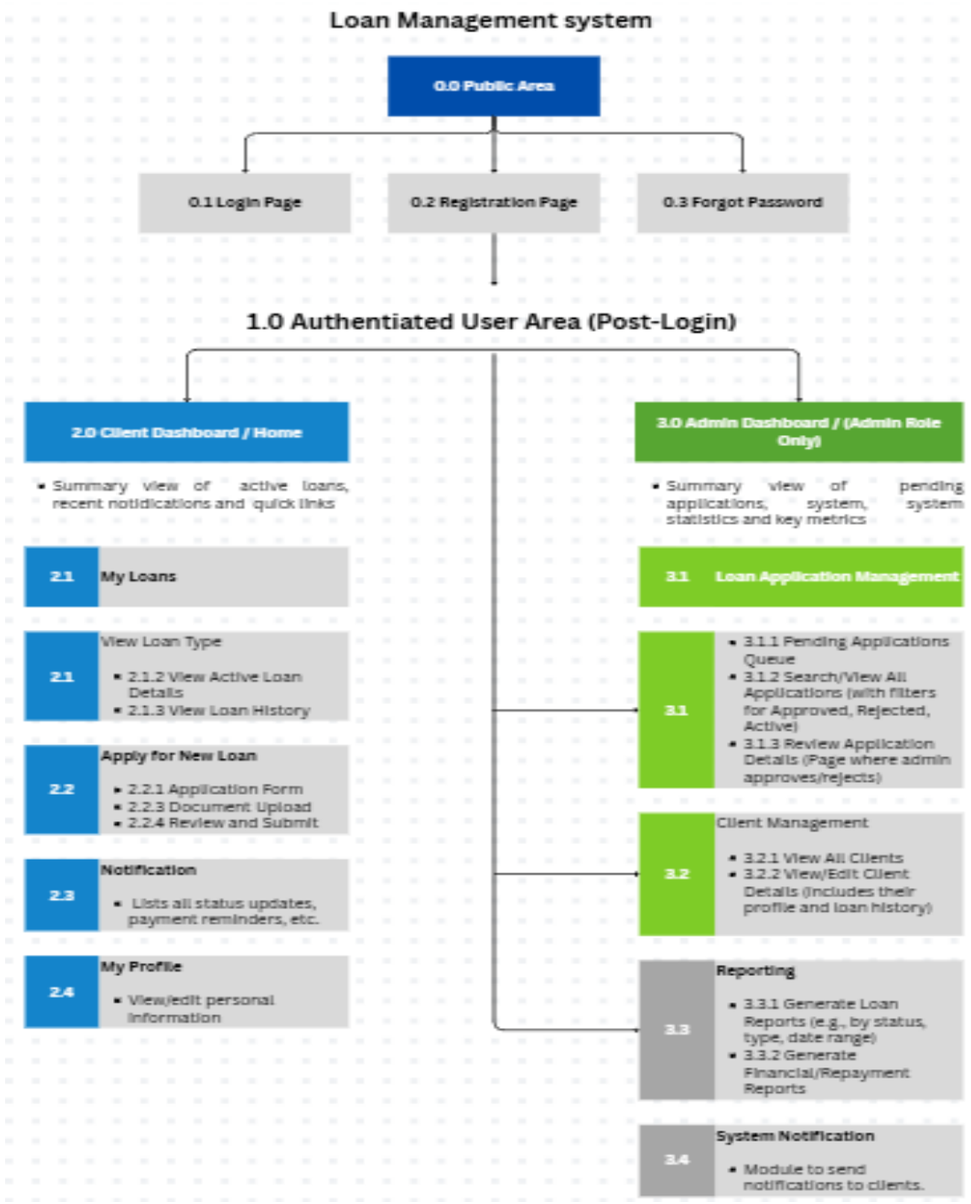
DFD Level 0 or Context Diagram



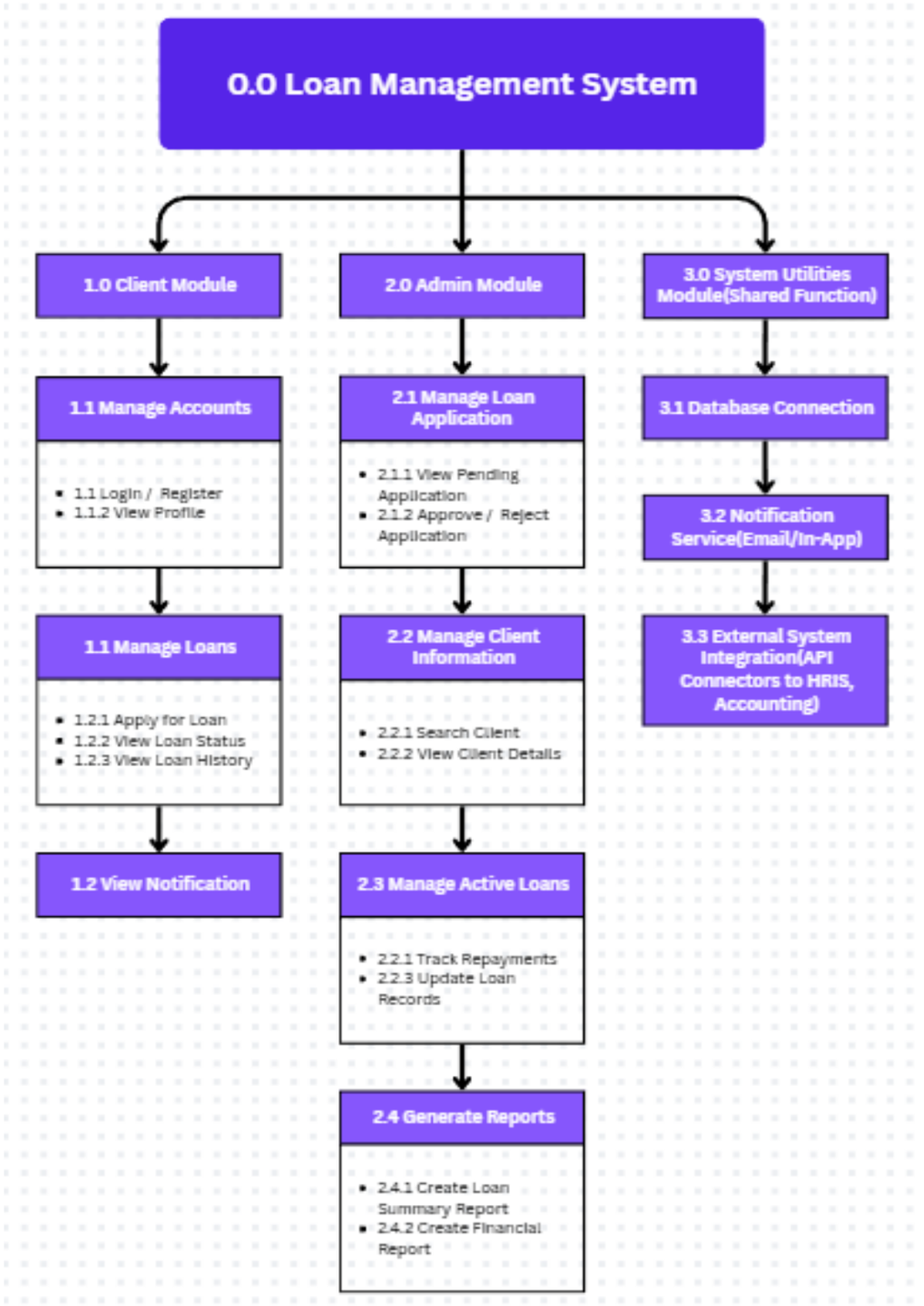
Activity Diagram



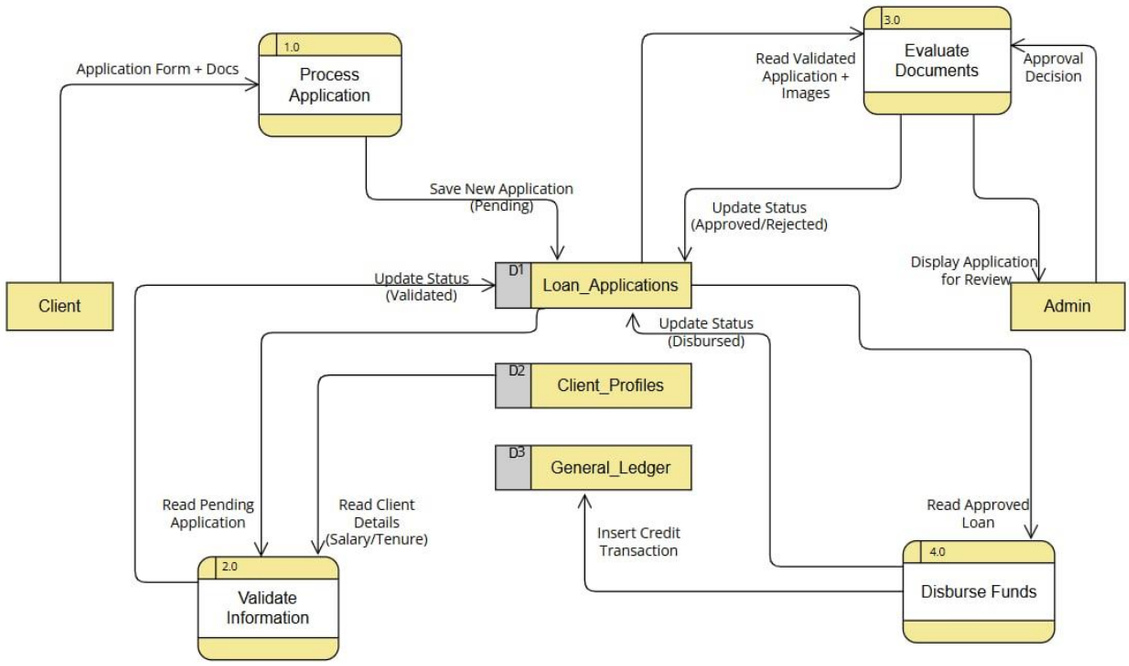
Site Map


Loan Management system
0.0 Public Area
0.1 Login Page
0.2 Registration Page
0.3 Forgot Password
1.0 Authentiated User Area (Post-Login)
2.0 Client Dashboard / Home
• Summary view of active loans, recent notidications and quick links
2.1 My Loans
2.1 View Loan Type
• 2.1.2 View Active Loan Details
• 2.1.3 View Loan History
2.2 Apply for New Loan
• 2.2.1 Application Form
• 2.2.3 Document Upload
• 2.2.4 Review and Submit
2.3 Notification
• Lists all status updates, payment reminders, etc.
2.4 My Profile
• View/edit personal information
3.0 Admin Dashboard / (Admin Role Only)
• Summary view of pending applications, system, system statistics and key metrics
3.1 Loan Application Management
3.1
• 3.1.1 Pending Applications Queue
• 3.1.2 Search/View All Applications (with filters for Approved, Rejected, Active)
• 3.1.3 Review Application Details (Page where admin approves/rejects)
3.2 Client Management
• 3.2.1 View All Clients
• 3.2.2 View/Edit Client Details (Includes their profile and loan history)
3.3 Reporting
• 3.3.1 Generate Loan Reports (e.g., by status, type, date range)
• 3.3.2 Generate Financial/Repayment Reports
3.4 System Notification
• Module to send notifications to clients.

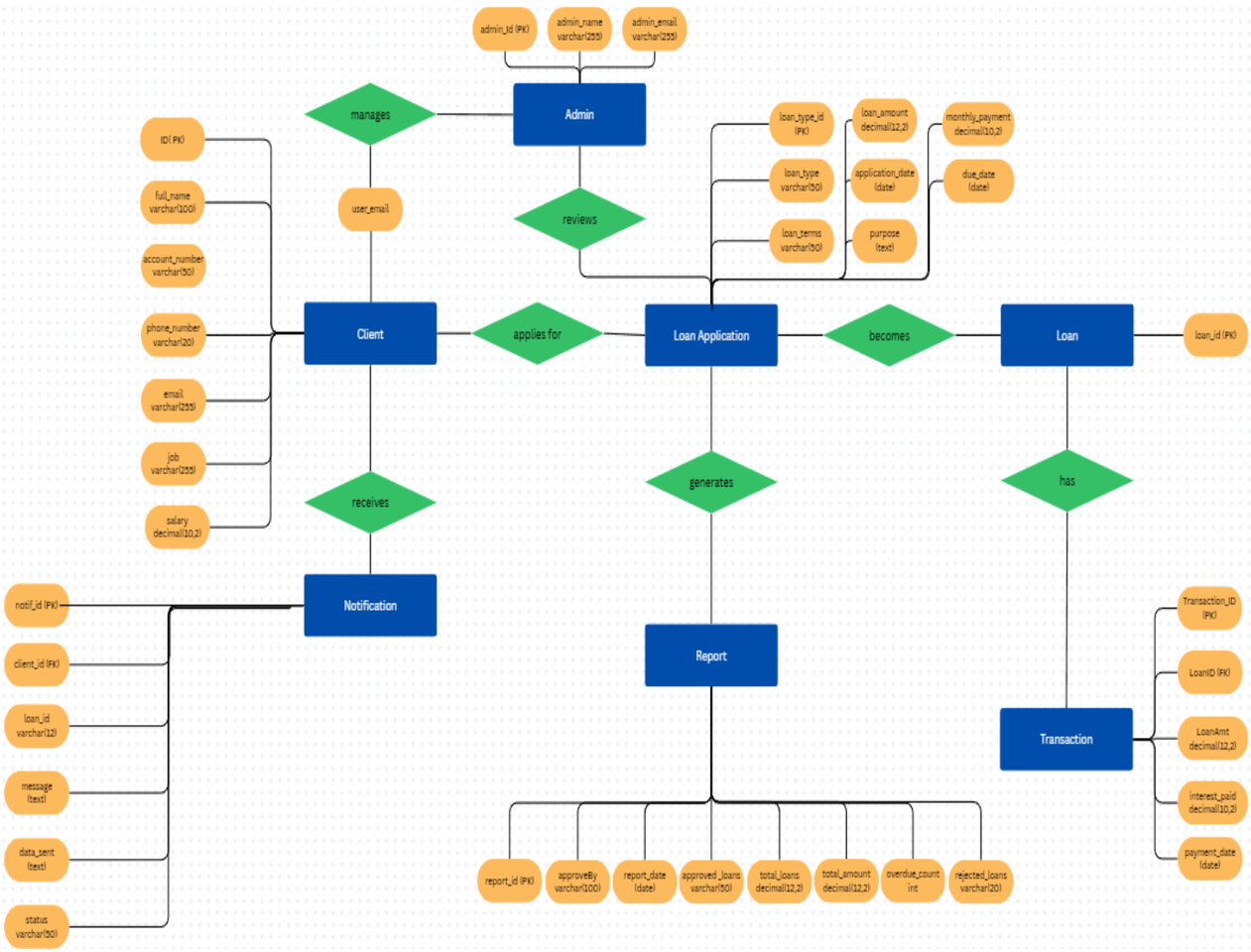
VTOC Diagram



Data Flow Diagram (Level 1)



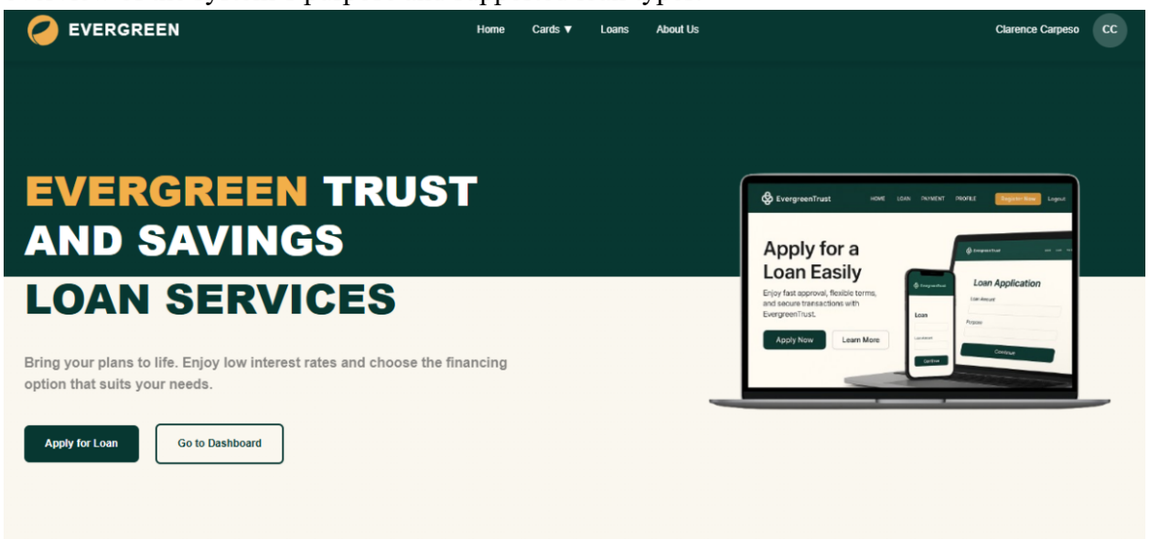
Entity Relationship Diagram



Screenshot with Description

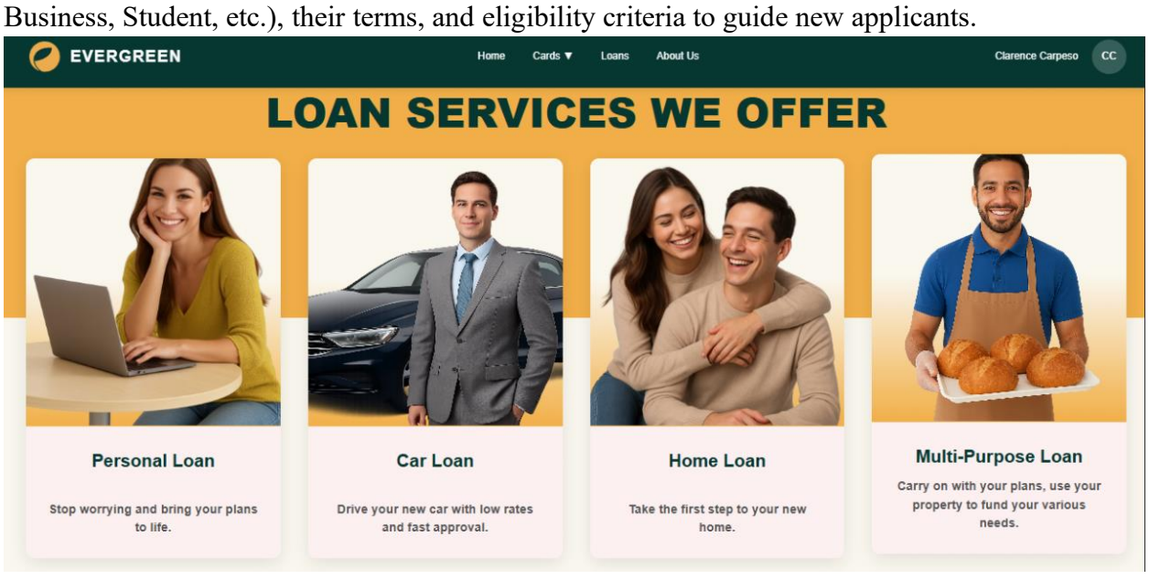
1. Landing page

The entry point for all users, featuring navigation to login or registration, and a brief overview of the system's purpose and supported loan types.



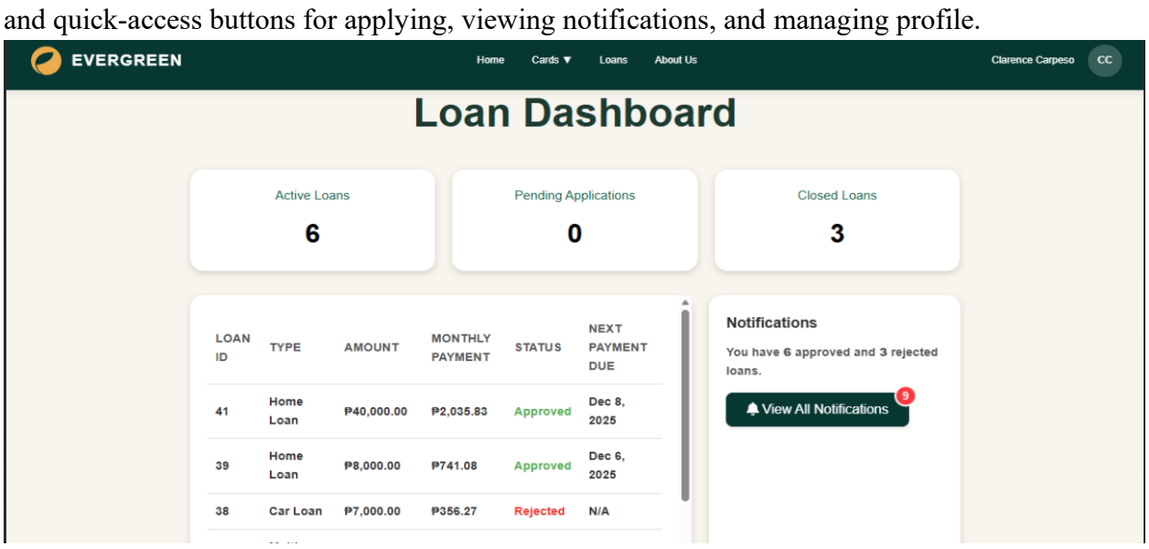
2. Loan Offer

A promotional or informational screen that highlights available loan categories (Personal, Business, Student, etc.), their terms, and eligibility criteria to guide new applicants.



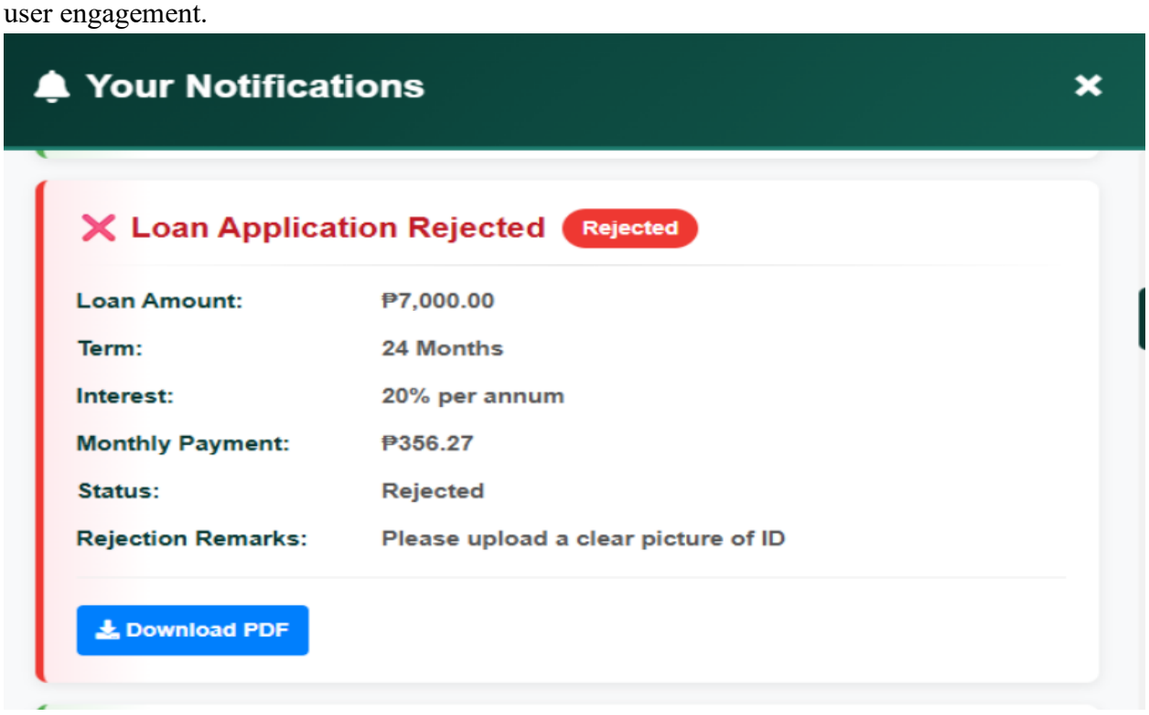
3. Dashboard

The central hub for logged-in clients, displaying current loan status, application history, and quick-access buttons for applying, viewing notifications, and managing profile.



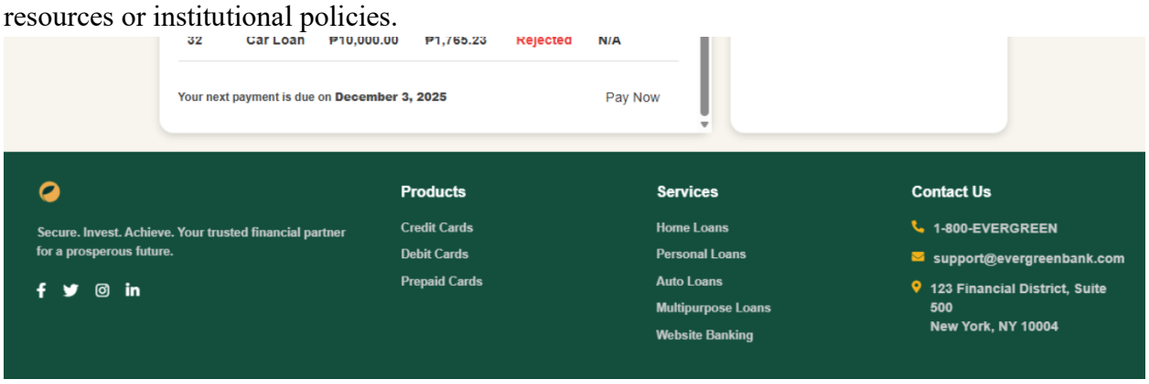
4. Notifications

A dedicated panel where clients receive real-time alerts including application approval, disbursement confirmation, due date reminders, and overdue notices enhancing transparency and user engagement.



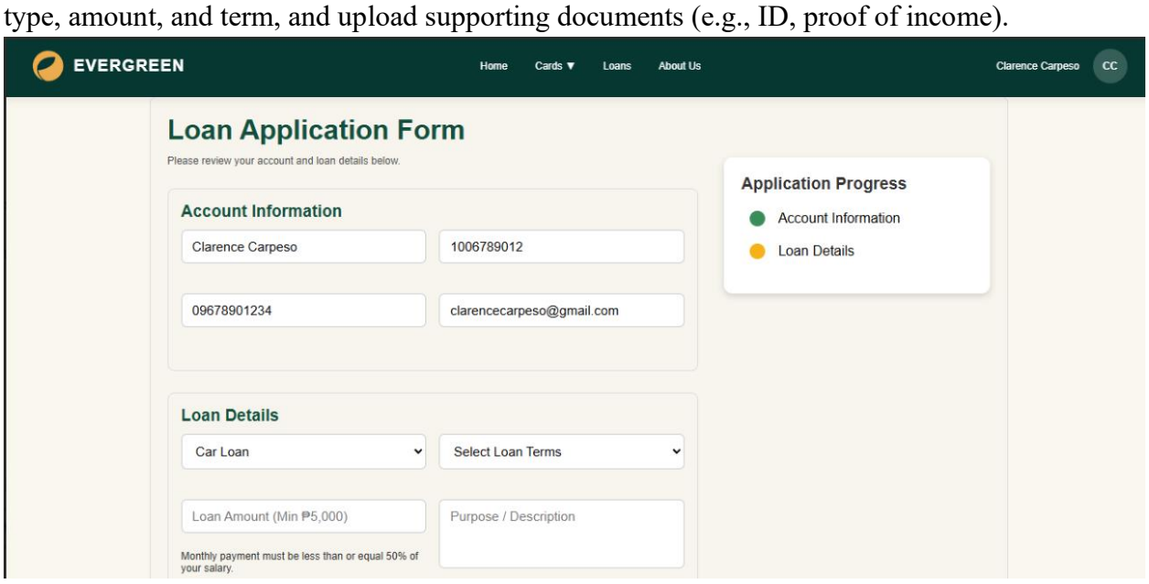
5. Footer

Standard footer section containing copyright, contact information, and links to help resources or institutional policies.



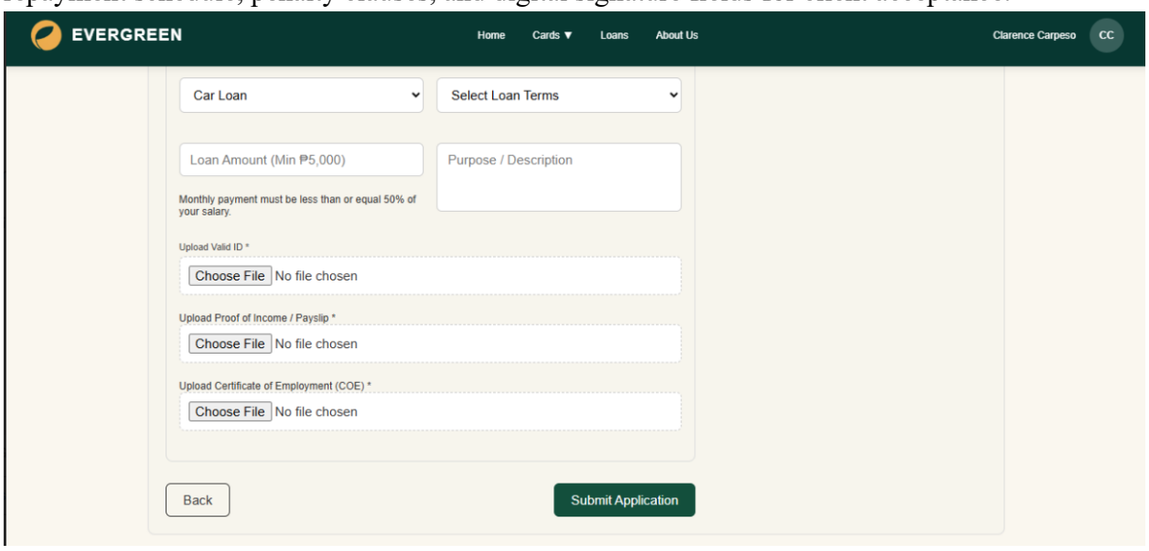
6. Loan Application Form #1

The first step in the application process, where clients input personal details, select loan type, amount, and term, and upload supporting documents (e.g., ID, proof of income).



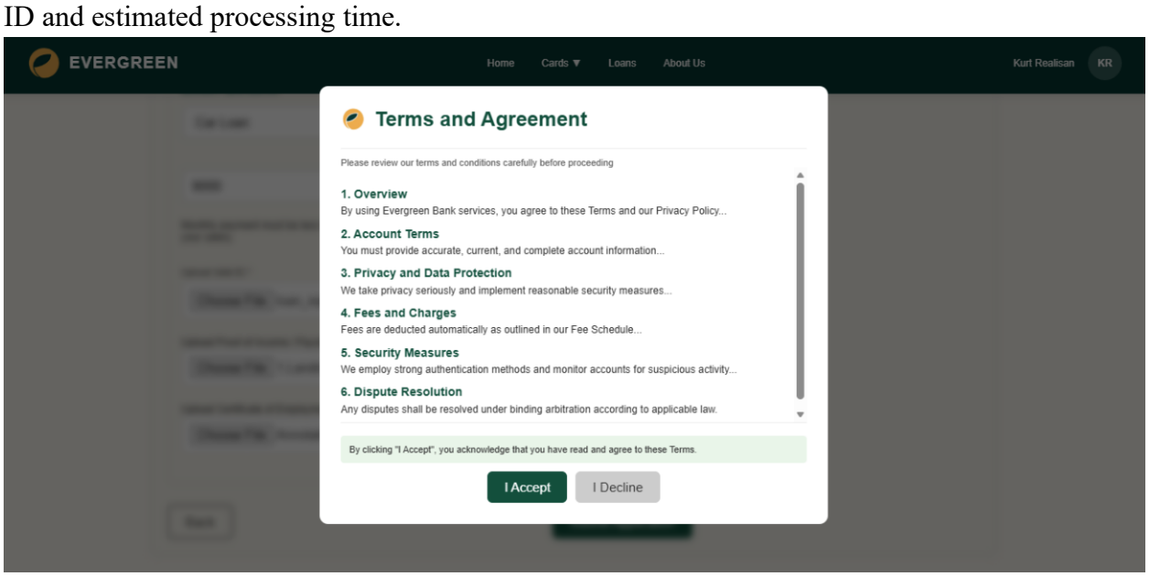
7. Loan Application Form #2

A legally binding screen presented after application submission, outlining interest rates, repayment schedule, penalty clauses, and digital signature fields for client acceptance.



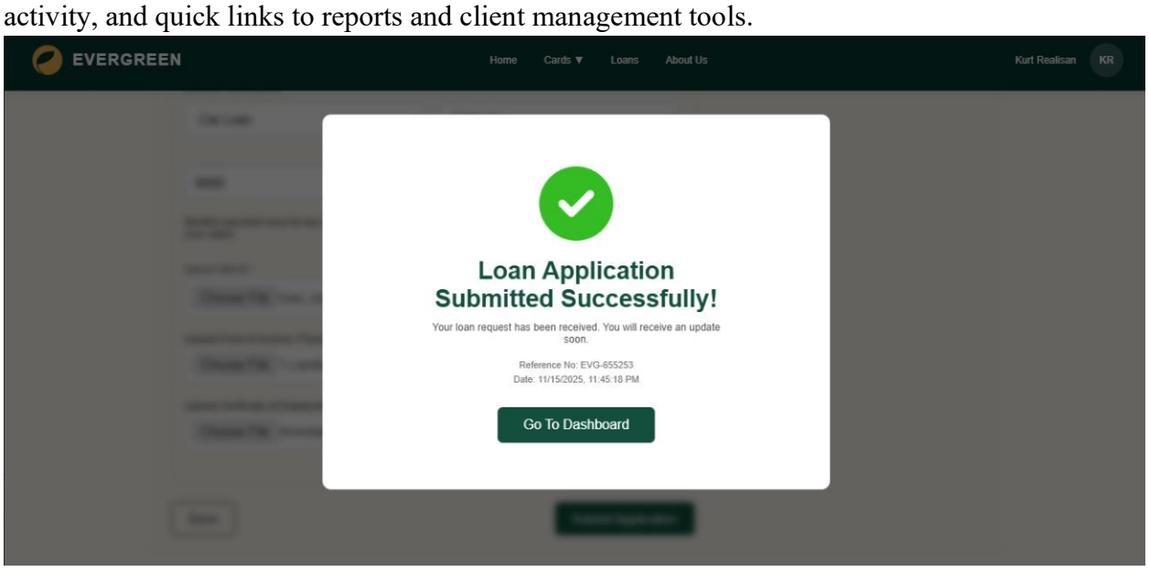
8. Terms & Agreement

A confirmation page displayed after successful form submission, providing an Application ID and estimated processing time.



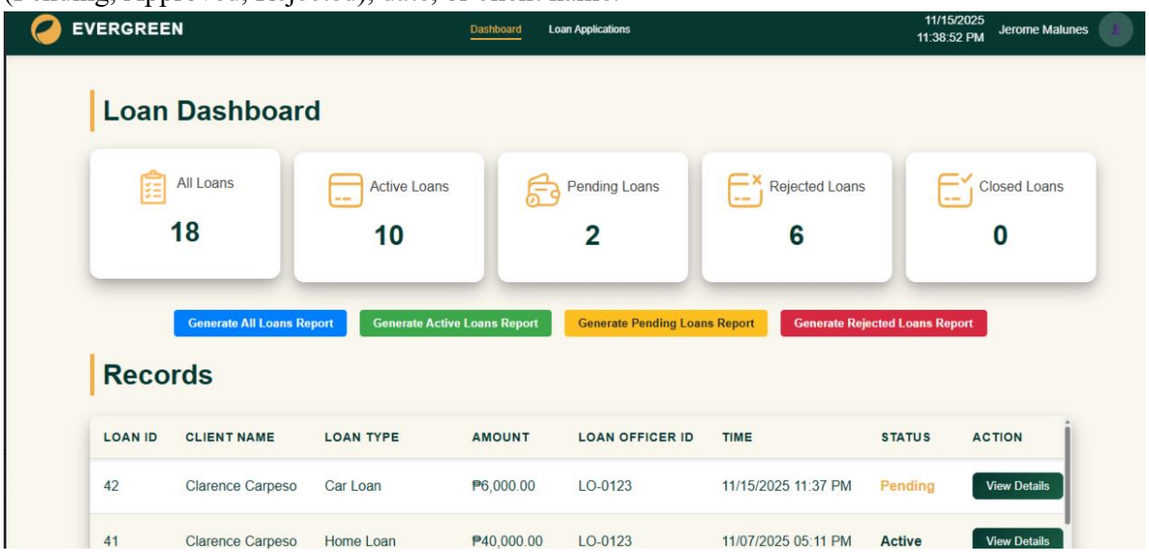
9. Submitted Successfully

The control center for administrators, offering an overview of pending applications, recent activity, and quick links to reports and client management tools.



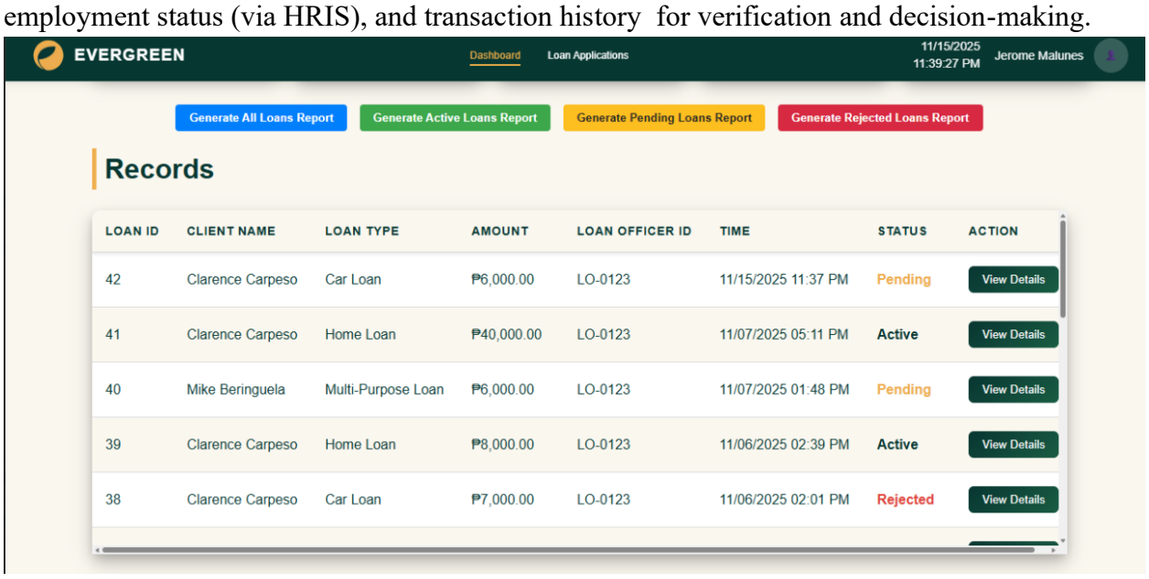
10. Admin Dashboard

A comprehensive list view of all loan applications, searchable and filterable by status (Pending, Approved, Rejected), date, or client name.



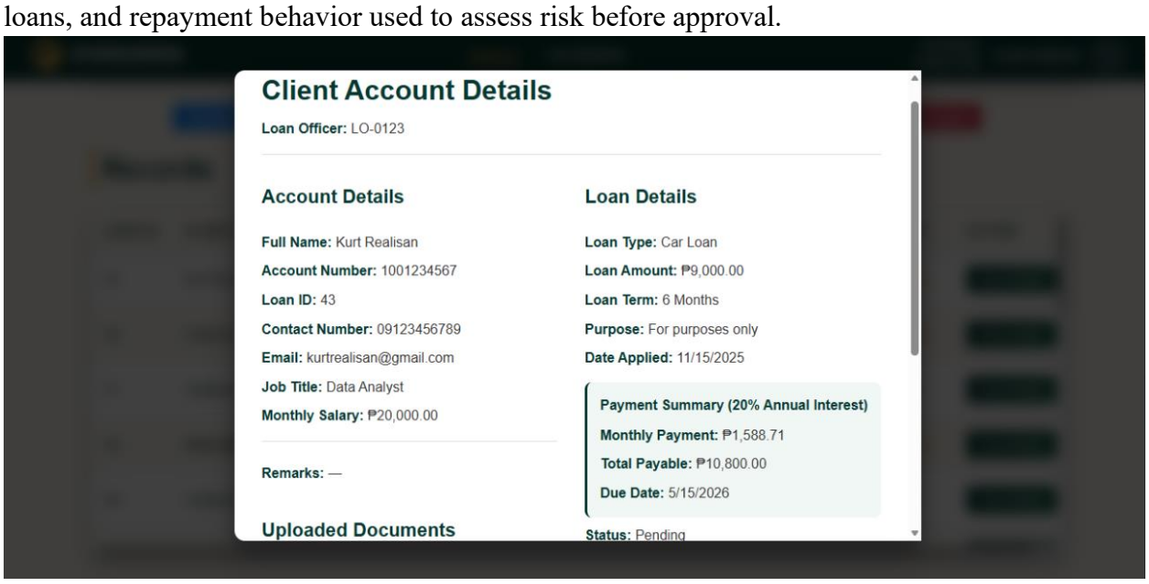
11. Admin Records

An admin-view profile page showing the client's full information name, contact, employment status (via HRIS), and transaction history for verification and decision-making.



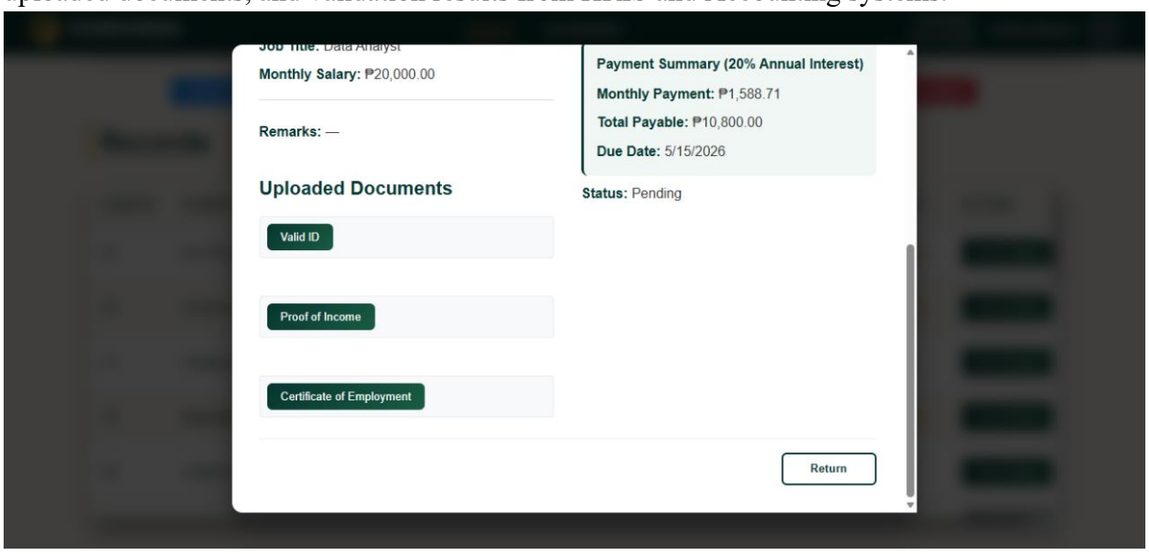
12. View Client Account Details #1

A deeper view of the client's financial standing, including credit score (if integrated), past loans, and repayment behavior used to assess risk before approval.



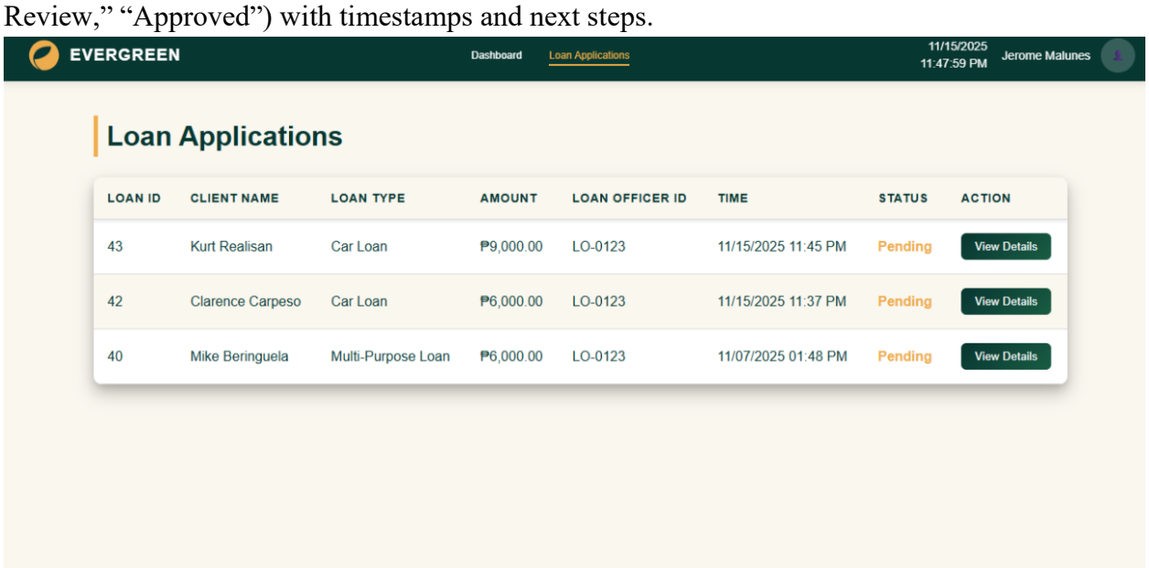
13. View Client Account Details #2

The detailed review interface for administrators, displaying the submitted application form, uploaded documents, and validation results from HRIS and Accounting systems.



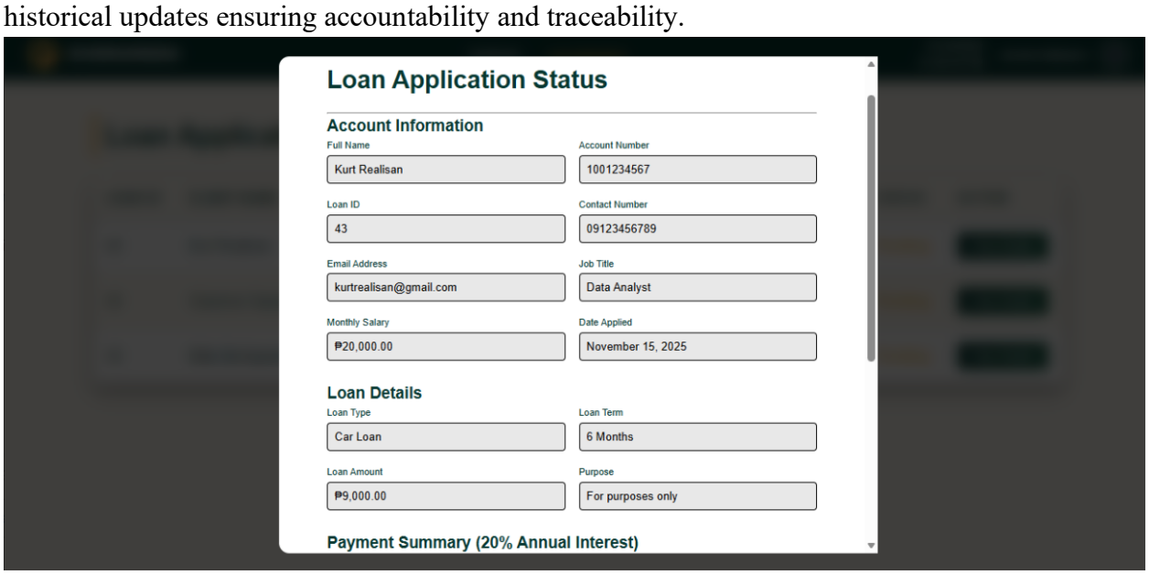
14. Loan Application (Admin)

A status tracker for clients, showing the current stage of their application (e.g., "Under Review," "Approved") with timestamps and next steps.



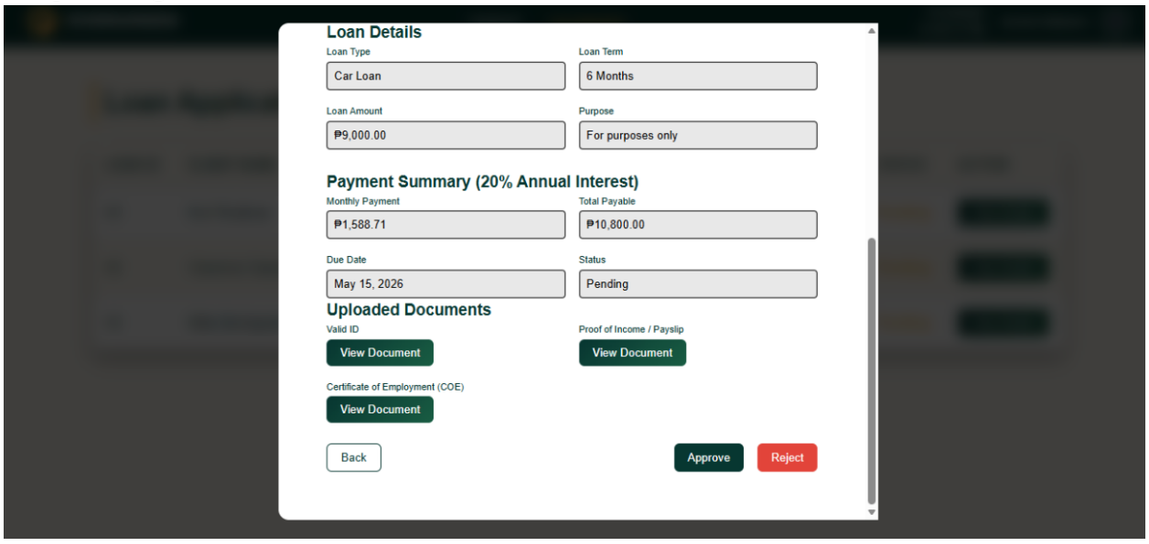
15. Loan Application Status #1

An expanded status view for admins, including audit logs, approver comments, and historical updates ensuring accountability and traceability.



16. Loan Application Status #2

Final status screen for both client and admin upon loan closure, confirming full repayment and providing a downloadable settlement certificate.



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Curriculum Vitae

 Beringuela, Jiro C. **Member** ***Project Manager***	 **Carpeso, Kurt Francis O.** **Member** ***Backend Programmer***

OBJECTIVE:
To secure an entry-level position or internship in Information Technology where acquired academic knowledge and technical skills can be applied and developed to contribute to organizational goals.

PERSONAL INFO
- **Nickname : Ro**
- **Age : 21**
- **Birthday : 07 July, 2004**
- **Birthplace : Quezon City**
- **Religion : Roman Catholic**
- **Contact no. : 0951-567-1234**
-

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023- Present

• Senior High school: Information Communication Technology (ICT)
Immaculate Conception Polytechnic
2021 - 2023

SPECIAL SKILLS & INT
Technical: HTML, CSS, JavaScript, PHP, XAMPP, phpMyAdmin, ASP.NET, C#, .NET Framework, Spring Boot
(Java), Oracle SQL, UI/UX Concepts
Creative: Digital Art, Storyboarding, Scriptwriting, Logo Design, Visual Layout
Soft Skills: Collaboration, Problem-Solving, Communication, Adaptability

REFLECTION
My studies in IT have focused on developing a strong foundation in both technical skills and the human element of computing. I am committed to continuous learning and eager to apply theoretical knowledge to solve real-world problems.

OBJECTIVE:
To build a strong foundation for future careers in SQL database management and cybersecurity, focusing on developing technical skills, secure data handling practices, and effective problem-solving in modern IT environments.

PERSONAL INFO
- **Nickname : Kurt**
- **Age : 20**
- **Birthday : 10 June, 2005**
- **Birthplace : Quezon City**
- **Religion : Roman Catholic**
- **Contact no. : 0995-9228310**

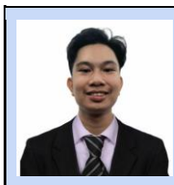
EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023- Present

•High school : Sauyo High School
2016-2020

SPECIAL SKILLS & INT
- **SQL Queries**
- **Basic Oracle & No SQL**
- **Basic Python & C#**

REFLECTION
I learned new programming skills and improved my understanding of databases and cybersecurity, helping me grow more confident as an IT student.



Edillor, Jestony
Member
Frontend Programmer

Gualva, Jerome M.
Member
Database

OBJECTIVE:
To gain knowledge in programming that can improve my skills and help my future career.

PERSONAL INFO
Nickname : Jes
Age : 21
Birthday : 29 Oct, 2004
Birthplace : Quezon City
Religion : Roman Catholic
Contact no. : 09686281447

EDUCATIONAL BACKGROUND

College : BS Information Technology
Quezon City University
Quezon City
2023- Present

High school : Caybiga Highschool
2021 - 2023

: Caybiga Highschool
2018 - 2021

: Caybiga Elementary School
2017 - 2018

SPECIAL SKILLS & INT
Basic React Programming
Basic Back end Programming
Adaptability

REFLECTION
In order for me to excell I need to explore outside of my comfort zone, from what I learned in this project is that I should not limit my self and should explore every technology that I can in order to become a better programmer.

OBJECTIVE:
To have a skillset that helps me in my future career.

PERSONAL INFO
- **Nickname : Jerome**
- **Age : 21**
- **Birthday : 6 September, 2004**
- **Birthplace : Quezon City**
- **Religion : Roman Catholic**
- **Contact no. : 09938761876**

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023- Present

•High school : San Francisco High School
2017 - 2023

SPECIAL SKILLS & INT
- **Basic knowledge in programming language (SQL, Java, Html, Css)**
- **Can communicate to colleagues**
- **Multi-tasking**

REFLECTION
I'm learning that technical skills alone aren't enough what truly matters is curiosity, the willingness to collaborate, and the discipline to keep improving. Every line of code, every team discussion, and every problem solved is building not just my competence, but my character as a future-ready IT professional."

Malunes, Angelo
Member
Frontend Programmer

Pinto, Larence
Member
UI/UX Designer

OBJECTIVE:
To improve my programming skills to become professional

PERSONAL INFO

- **Nickname : Zion**
- **Age : 21**
- **Birthday : 13 Oct, 2004**
- **Birthplace : Quezon City**
- **Religion : Roman Catholic**
- **Contact no. : 0992-727-5667**

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023 - Present

• Senior High School :
Immaculada Concepcion College
2021-2023

•High school : Manuel Luis Quezon High School

2017-2021

SPECIAL SKILLS & INT

- **Basic Coding**
- **Creative Designing**
- **Critical Thinking**
- **Fast Learner**

REFLECTION
I've recently picked up a new programming language, and now I'm eager to apply my skills to create innovative designs that challenge me and help me grow.

OBJECTIVE:
To improve my programming skills, front end development and communication skills in order to become a better representative.

PERSONAL INFO

- **Nickname : Yence**
- **Age : 21**
- **Birthday : 28 Oct, 2004**
- **Birthplace : Quezon City**
- **Religion : Roman Catholic**
- **Contact no. : 0997-880-9937**

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023 - Present

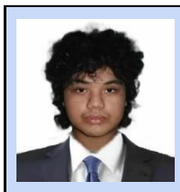
• Senior High School : Saint John Of Beverly
2021-2023

•High school : Kings Montessori School
2017-2021

SPECIAL SKILLS & INT

- **Basic Coding**
- **UI / UX front end development**
- **Creative Designing**
- **Critical Thinking**
- **Fast Learner**

REFLECTION
I've learned how to be resourceful, and to adapt in any circumstances my team has given me. I've found out that front end development is my forte and I am eager to improve my skills more as I keep on learning and adapting.



Realisan, Mike L.
Member
System Analyst

Tordillos, Clarence
Member
Backend Programmer

OBJECTIVE:
To gain knowledge in programming that can improve my skills and help my future career.

PERSONAL INFO

- **Nickname : Miky**
- **Age : 21**
- **Birthday : 3 November, 2004**
- **Birthplace : Paranaque City**
- **Religion : Roman Catholic**
- **Contact no. : 09275344737**

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023- Present

•Senior High school : Emilio Jacinto National Highschool
2021 - 2023

•High school : Sauyo High School
2018 - 2021

SPECIAL SKILLS & INT

- **Basic knowledge in programming language (SQL, Java, Html, Css)**
- **Critical Thinking**
- **Multi-tasking**

REFLECTION
Growth rarely happens inside the comfort zone especially in programming. Every time I push through a confusing bug, try a new concept, or refactor messy code, I'm stretching beyond what's familiar. I'm learning to embrace the discomfort of not knowing, because that's where real skill and confidence is built.

OBJECTIVE:
To enhance my skills for both front-end and backend because I'm aspiring full-stack developer.

PERSONAL INFO

- **Nickname : Clarence**
- **Age : 20**
- **Birthday : 23 July, 2005**
- **Birthplace : Manila**
- **Religion : Christian**
- **Contact no. : 09369689137**

EDUCATIONAL BACKGROUND

•College : BS Information Technology
Quezon City University
Quezon City
2023- Present

•Senior High school : Lagro High School
2021 - 2023

•High school : North Fairview High School
2017 - 2021

SPECIAL SKILLS & IN IT

- **FrontEnd**
- **BackEnd**
- **Database**

REFLECTION
I'm always pushing my limits to learn more because I'm preparing myself for the industry. I need to practice more to build a portfolio, foundation, and network.